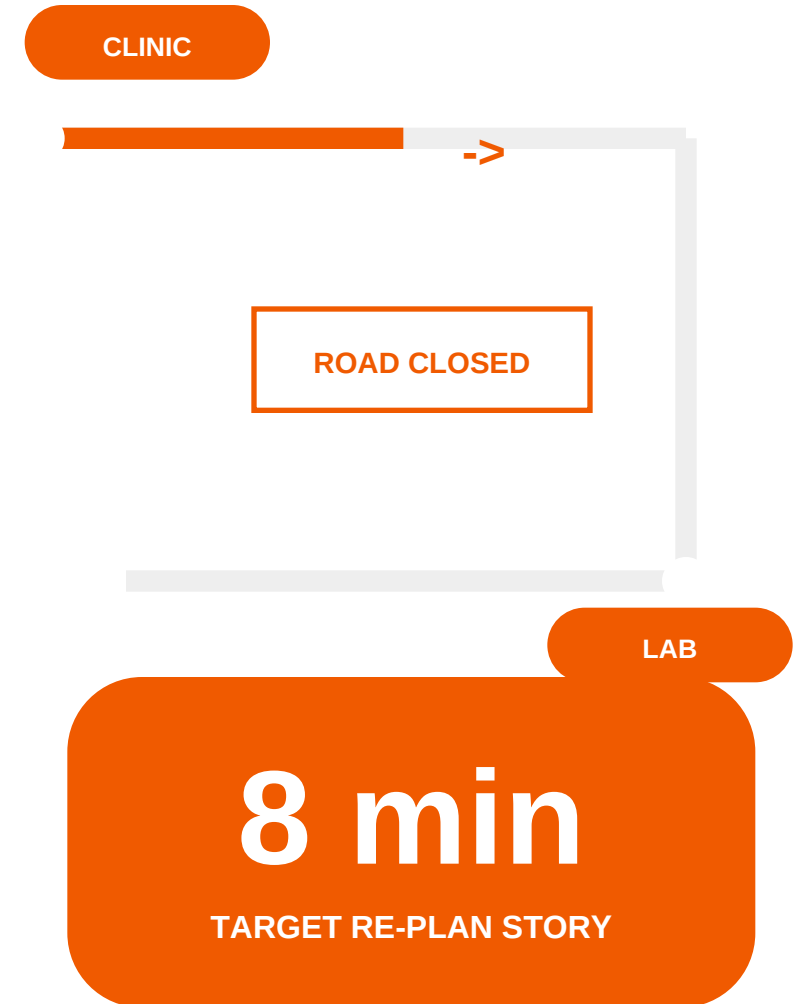


HACK FOR IMPACT LONDON | NVIDIA

RLJ

Time-critical medical logistics for London.

- Urgent samples, medicines, couriers and road disruption in one live dispatch loop.
- Built for a local DGX Spark: patient data stays on the box.

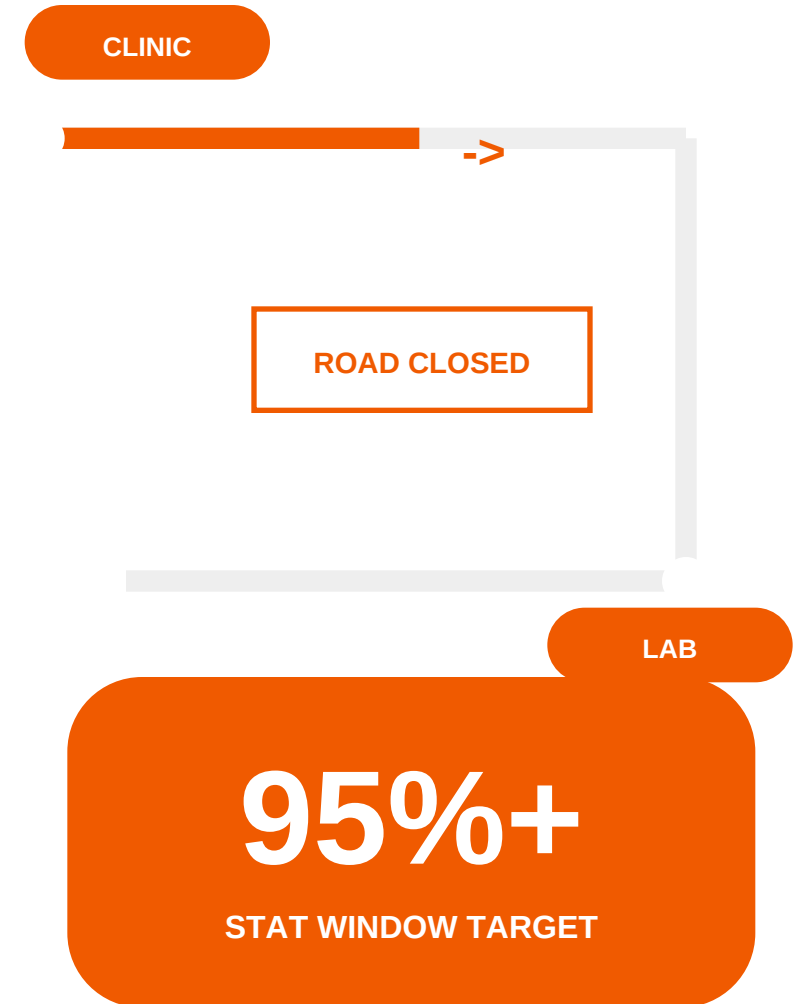


THE PROBLEM

Clinical windows do not wait for traffic.

London medical couriers operate inside constraints that ordinary routing tools ignore.

- STAT samples have stability windows.
- Courier capacity changes minute by minute.
- Closures, congestion and disruptions break static plans.
- Phones still matter: clinics and couriers need clear ETAs.

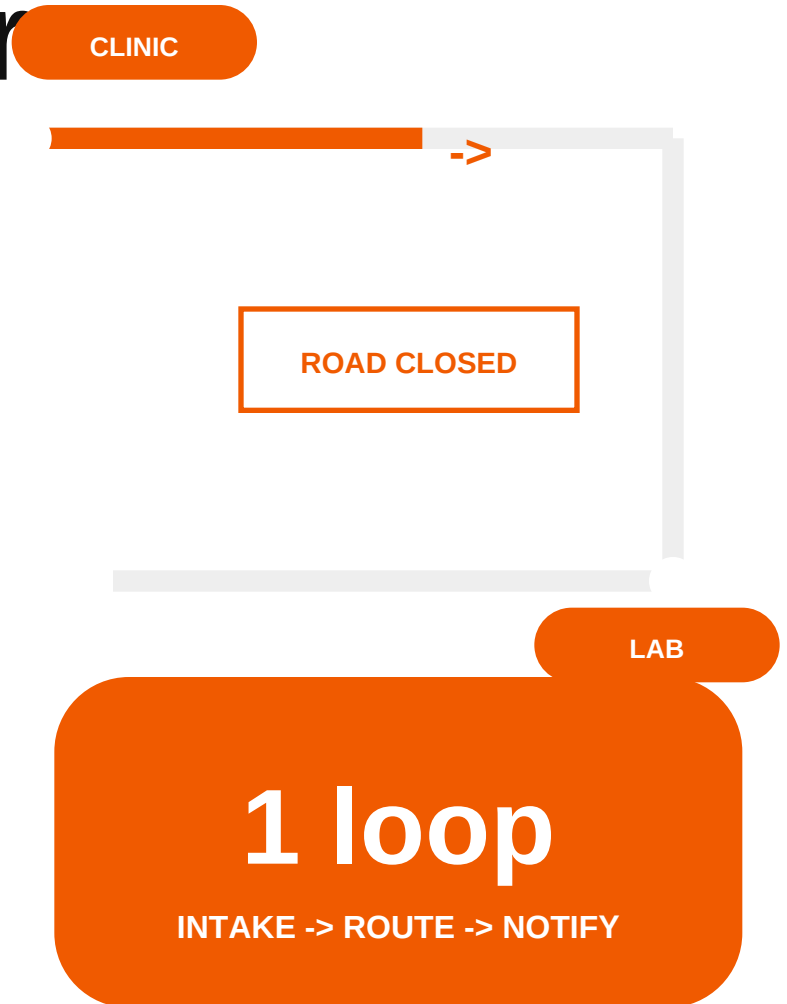


THE ANSWER

A local command center for urgent healthcare delivery.

RLJ turns calls and live city signals into dispatch decisions, then explains them.

- Voice intake creates structured delivery jobs.
- Orchestrator keeps jobs, couriers, plans and events in sync.
- Routing re-optimizes when the city changes.
- Frontend shows the decision, not just the map.

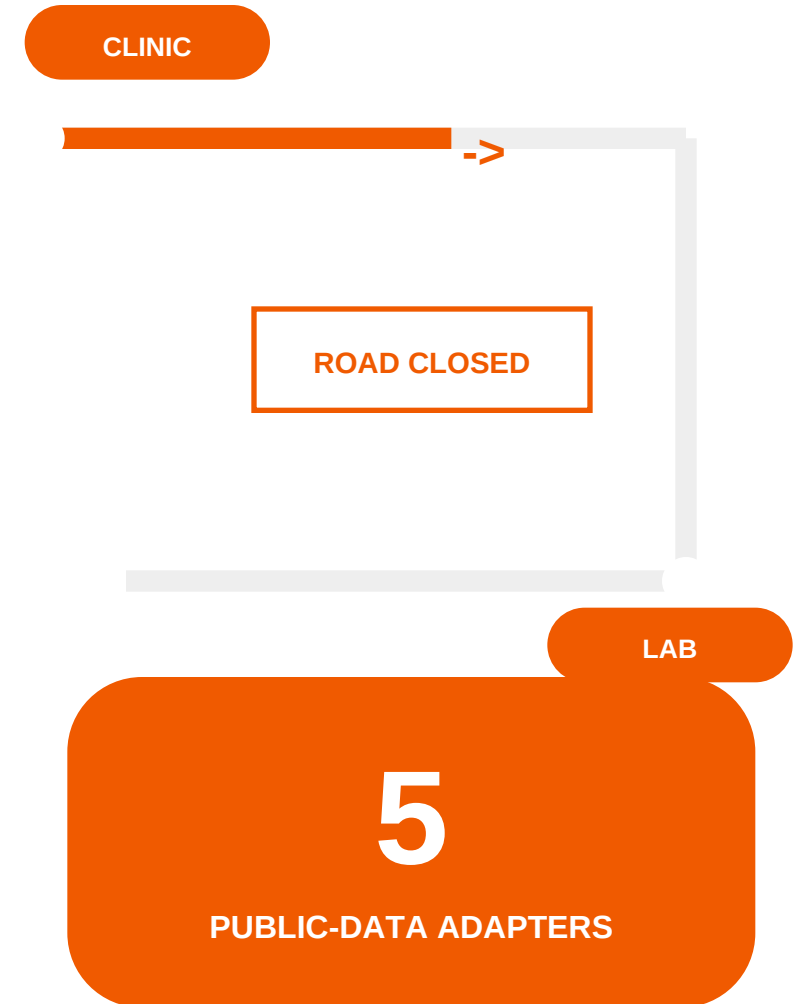


LONDON DATA LAYER

Live city context, normalized into safe local records.

The demo is not a canned map. It can ingest London open data and turn it into operational signals.

- TfL road disruptions -> route risk and closures.
- TfL BikePoint -> staging-point metadata.
- TfL line status -> city mobility risk narration.
- LondonAir + London Datastore -> public context and dataset discovery.

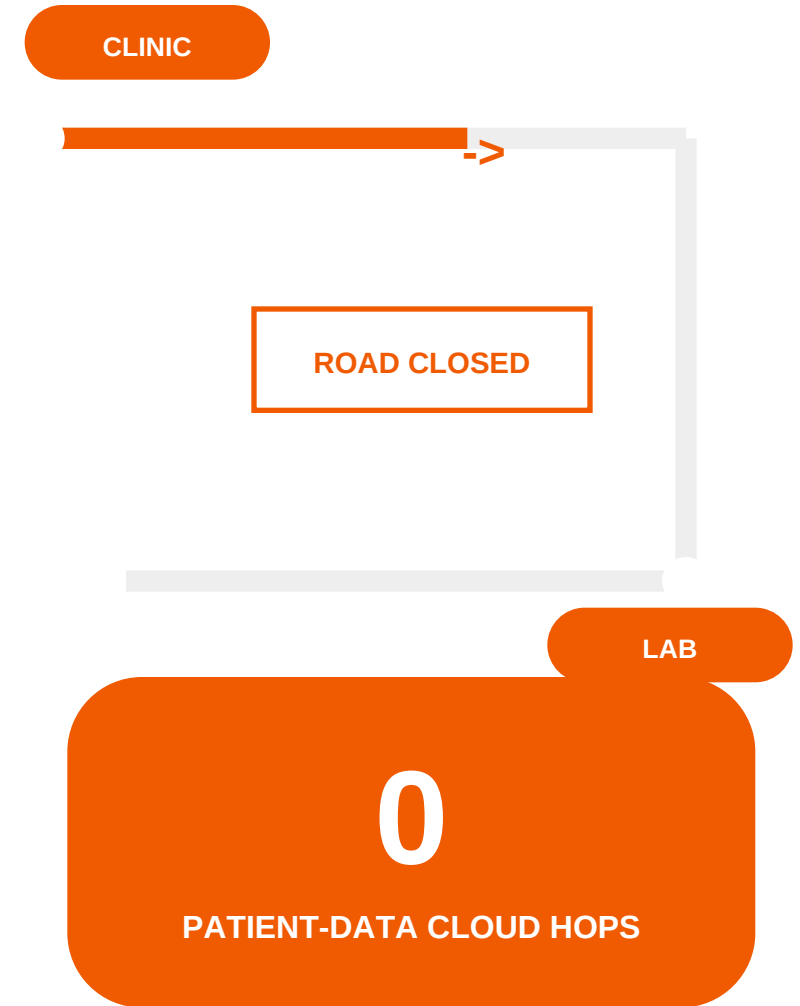


LOCAL-FIRST ARCHITECTURE

Reason locally. Route locally. Reveal only what is needed.

DGX Spark is the story: low latency, privacy, resilience and repeatable routing.

- Local Nemotron/NemoClaw reasoning and policy control.
- GPU routing with a built-in greedy fallback for demo safety.
- Orchestrator exposes REST + WebSocket contracts.
- ElevenLabs and TfL are allowlisted edges, not the core dependency.

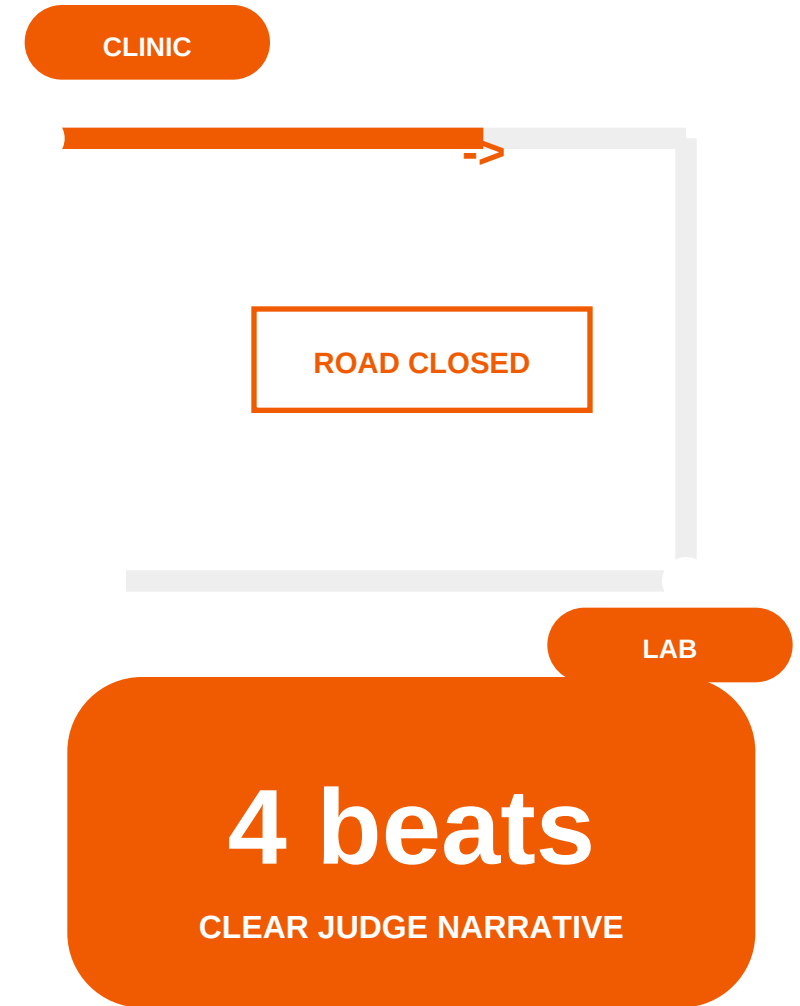


DEMO PLAN

Show the city breaking the plan, then RLJ fixing it live.

The winning moment is visual, audible and measurable.

- 1. Clinic calls in an urgent sample.
- 2. RLJ creates the job and computes the plan.
- 3. A road disruption lands from demo control or TfL.
- 4. RLJ re-routes, updates the map and sends a voice notification.

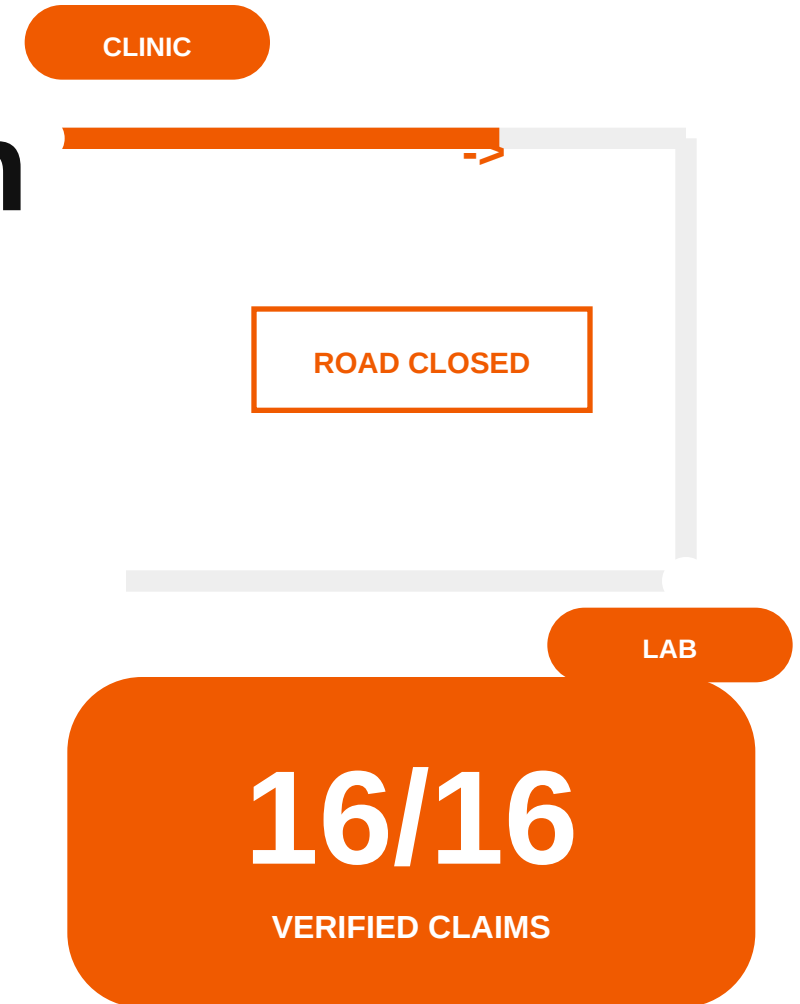


VERIFICATION

No self-grading. Claims only count when tests pass.

The project now has objective gates for the story it tells on stage.

- make quality-gate runs the Python verification ledger.
- Frontend TypeScript/Vite build is part of the gate.
- GitHub Actions repeats the gate for PRs.
- Current ledger: 14/14 must-pass checks green, 16/16 claims verified.



WHY IT CAN WIN

It feels like a real operating system for London healthcare.

Judges remember concrete pressure: a patient-critical delivery, a blocked road, a local AI decision, a live ETA call.

- High-impact problem with obvious urgency.
- NVIDIA/local-first angle is central, not decorative.
- Live data makes the demo feel connected to London.
- Quality gates make the pitch credible under questioning.

